



## Collect Speed Samples Data In Excel Format Using MATLAB

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**Y**ou can now get samples of real-time speed of a DC motor in MS Excel format using a MATLAB Graphical User Interface (GUI). The variations in motor's speed are also plotted automatically at the end of data acquisition. The GUI communicates with an Arduino that is interfaced with the speed sensor. The authors' prototype of the project is shown in Fig. 1.

The major components used and their function in this project are described below.

**Speed sensor.** The opto-coupler type speed sensor module used here generates square wave of a specific frequency, which is a function of the speed of motor. Opto-coupler module FC-03 was used during testing. The internal circuit diagram of this speed sensor is shown in Fig. 2.

**Coded disc encoder.** It has a certain number of holes that help the speed sensor to generate a square wave of specific frequency. A 20-hole coded disc encoder was used for prototype.

**Arduino Uno board.** Arduino Uno

is an AVR ATmega328P microcontroller based development board with 6 analogue and 14 digital I/O pins. The microcontroller has 32kB ISP flash memory, 2kB RAM, and 1kB EEPROM. It can operate at a clock frequency of 16MHz. The board provides capability of serial communication via UART, SPI, and I2C. Its digital pin no. 2 is used for detecting pulses from the speed sensor.

The square wave generated by the sensor module is available at digital pin no. 2 (interrupt pin INT0) of the Arduino. The DC motor's speed is checked every ten seconds. Since the disc has 20 holes, the speed (in rpm) =  $(\text{pulse} \times 6) / 20$ . This calculation is performed by the Arduino, which then sends the sensor data to the GUI. The sensor-Arduino interfacing is shown in Fig 3.

### Software

Two different software programs are used in this project for the Arduino and the GUI.

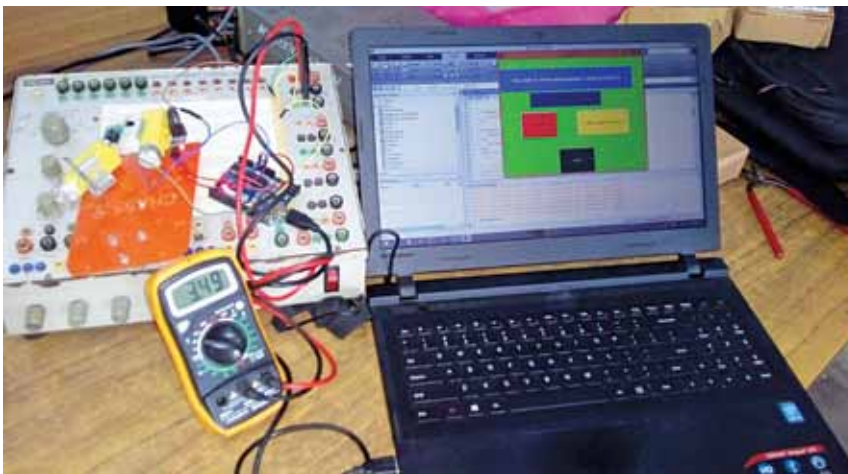


Fig. 1: Authors' prototype for speed sensing

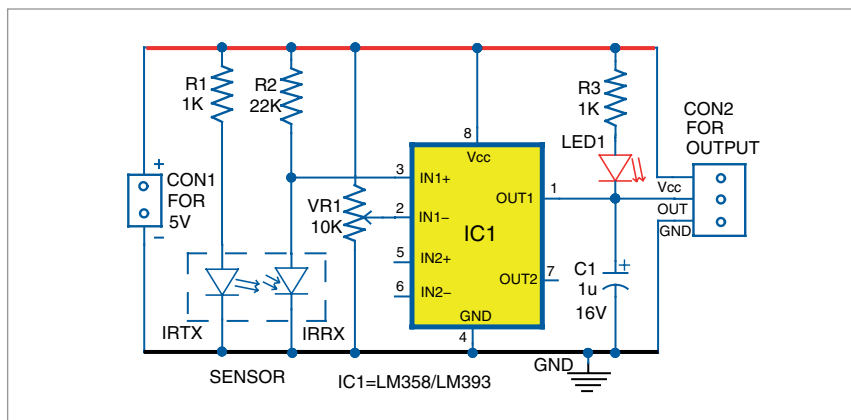


Fig. 2: Internal circuit of opto-coupler type speed sensor

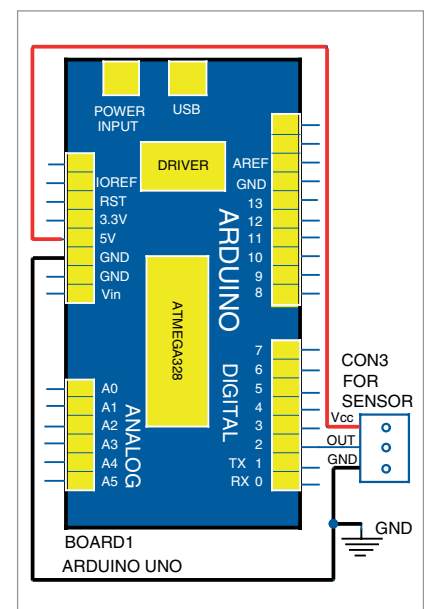


Fig. 3: Interfacing Arduino with sensor